PAGE 1 — IDENTITY AND CONCEPT

My name is MANZI PRINCE Peter

CMPortfolio

Personal Portfolio Mobile Application

Course: SWE 3409 — Mobile Application Development
Institution: Institut d'Enseignement Supérieur de Ruhengeri (INES-Ruhengeri)
Assessment: Individual Assignment 2
Student: MANZI PRINCE Peter
Registration Number: 25/28149
Verification Code: MOB-A2-8149
Platform: Android — React Native, Expo and TypeScript

1. Project Overview

CMPortfolio is a personal portfolio mobile application designed to introduce my professional identity, technical abilities, educational background and software projects through a clear and engaging mobile experience.

The application transforms a conventional CV into an interactive portfolio that a recruiter can explore directly on an Android device. It is designed to operate offline for its core portfolio features, with editable profile information and locally saved profile media.

2. Design Problem

A traditional CV provides limited opportunities to demonstrate interactive mobile development, interface design and practical software engineering skills. CMPortfolio addresses this limitation by presenting personal information, skills, education and projects in an organized application with dedicated project-detail screens.

3. Intended Users

The primary users are internship recruiters, lecturers, project evaluators and professionals reviewing my technical background. The interface should allow a first-time user to understand my identity, skills and project experience within two minutes.

4. Visual Direction

The design uses a premium navy, white and gold visual identity. Midnight navy communicates professionalism and confidence, white provides clarity, and gold highlights important actions and selected interface elements.

The layout uses generous spacing, clear typography, rounded cards and a consistent navigation structure. The visual language is intended to be elegant without distracting from the portfolio content.

5. Core Design Principles

- **Clarity:** Important identity information appears immediately.
- **Consistency:** Shared typography, colors, spacing and reusable components.
- **Accessibility:** Readable contrast, descriptive labels and support for larger system text.
- **Reliability:** Predictable validation, clear feedback and local data persistence.
- **Credibility:** Projects describe genuine work and clearly identify my personal contribution.

PAGE 2 — USER FLOW AND INFORMATION ARCHITECTURE

1. Information Architecture

The application uses bottom-tab navigation for its major sections and a nested stack route for project details.

Primary navigation

- **Home:** Introduction, profile image, headline, biography and verification code.
- **Skills:** Technical skills and education or training timeline.
- **Projects:** Portfolio project cards and access to detailed project information.
- **Profile:** Editable professional information and profile-image controls.
- **About:** Student identity, application version and data-use statement.

The exact number and names of visible tabs may be adjusted to match the implemented navigation. The About screen must remain accessible.

2. User Flow

Launch application → Home → Explore portfolio → Open project details → Return to Projects

A second flow supports profile management:

Home or Profile → Edit profile → Validate fields → Save changes → Persist locally → Restart application → Restore saved profile

The media flow is:

Profile → Choose Camera or Gallery → Request permission → Capture or select image → Preview → Confirm replacement or cancel → Save local image reference

The error-recovery flow is:

Invalid input or denied permission → Display clear feedback → Correct input or open settings → Retry without losing existing profile data

3. State and Data Management

Profile editing uses controlled React Native inputs. Validation messages appear beside the relevant fields. Invalid data must not be saved.

AsyncStorage stores low-risk editable profile information and the selected local image reference. The application restores saved information when reopened. A reset action should allow the user to return to the initial profile state.

4. Offline Behavior

The Home, Skills, Projects, Project Details, Profile Editor and About screens should remain usable without an internet connection. Core portfolio information and locally stored images must not depend on a remote server.

Repository links and any optional external demonstrations must be clearly identified as requiring an internet connection.

5. Navigation and Recovery

Project details receive a stable project identifier through the stack route. Android back navigation returns to the previous project list without unnecessarily resetting the current session.

Permission denial and cancellation must preserve the existing profile image and other saved profile information.

PAGE 3 — ANNOTATED WIREFRAMES: HOME, SKILLS AND PROJECTS

1. Home Screen

Purpose: Communicate identity and professional direction immediately.

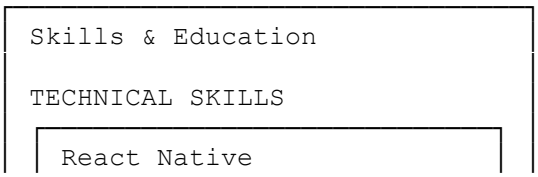


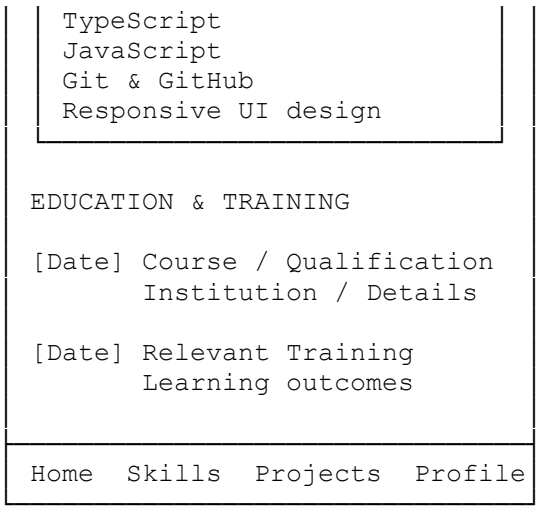
Annotations

1. The identity sentence is the first sentence displayed on the Home screen and uses the exact required wording.
2. The profile image uses a locally available photograph or student-created avatar.
3. The professional headline identifies my intended professional direction.
4. The biography briefly introduces my interests, current learning and goals without unsupported claims.
5. The verification code is visibly presented for assessment.
6. Bottom navigation provides direct access to the main portfolio sections.

2. Skills and Education Screen

Purpose: Present technical abilities and educational development in a scannable format.



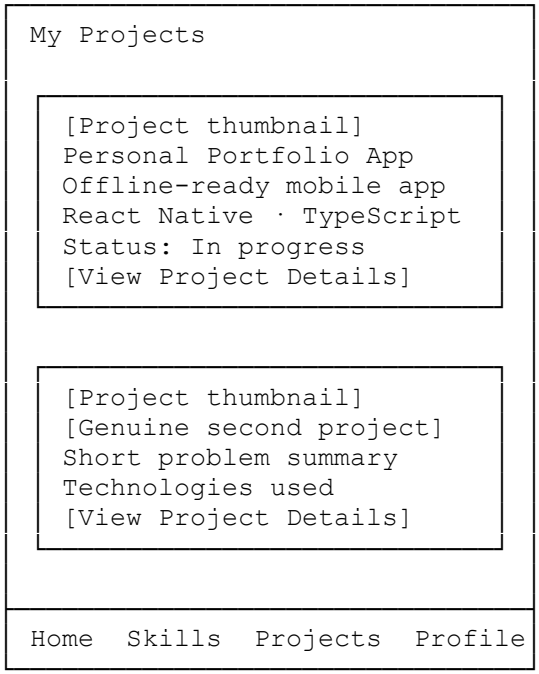


Annotations

- Skill entries use reusable components for consistent spacing and styling.
- Skills must represent technologies or abilities I genuinely possess.
- The education timeline displays real qualifications, coursework or training.
- Dates and descriptions should be concise and readable on narrow devices.
- Empty or unavailable information should be handled gracefully.

3. Projects Screen

Purpose: Demonstrate practical work and make each project easy to explore.



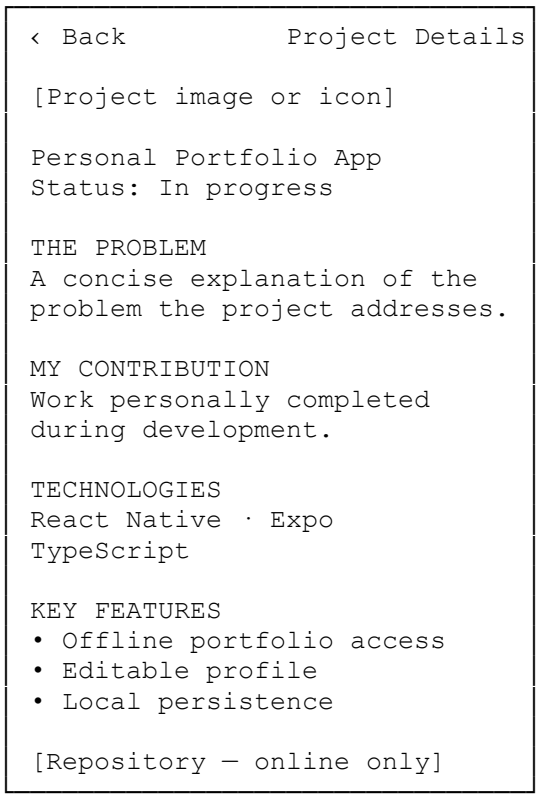
Annotations

- Each project card shows a title, short summary, technologies and status.
- The primary action opens a dedicated Project Details screen.
- The two required projects must be genuine student projects.
- Project thumbnails are optional; a polished placeholder or icon can be used when no suitable image exists.
- Cards must remain readable with larger system text and must not rely on color alone to communicate project status.

PAGE 4 — ANNOTATED WIREFRAMES: PROJECT DETAILS AND PROFILE EDITOR

1. Project Details Screen

Purpose: Explain the problem, personal contribution and technical implementation of an individual project.



Annotations

- The stack route receives a project identifier and displays the corresponding project.
- The Back action returns to the previous screen.
- The problem and personal contribution are separate to make the student's work clear.
- External repository or demonstration links are optional and explicitly labelled as requiring connectivity.
- Descriptions must accurately reflect the implemented project.

2. Profile Editor Screen

Purpose: Allow controlled editing of professional information with understandable validation.



Annotations

- All text fields are controlled by application state.
- Field-level messages explain how to correct invalid input.
- Saving is blocked until all required validation rules pass.
- Image selection and capture show previews before the user keeps the change.
- Removing or replacing a photo does not silently discard other profile information.

3. Required Validation and Error States

The implementation should include at least three meaningful validation rules.

Field	Example rule	Example feedback
Headline	Required; 3–60 characters	Enter a headline between 3 and 60 characters.
Biography	Required; 20–300 characters	Add a short biography of at least 20 characters.
Primary skill	Required; 2–40 characters	Enter a valid primary skill.

These are proposed validation rules for the interface; the implementation must enforce them before saving.

A validation error should appear directly below its field. Invalid fields should have a visible border or other clear indicator. Existing valid information must remain available while the user corrects errors.

4. Empty, Permission and Save States

- **No profile image:** Display a neutral avatar and image-selection actions.
- **Gallery cancelled:** Close the selection flow without changing the current image.
- **Permission denied:** Explain the missing permission and offer access to device settings where appropriate.
- **Saving:** Provide feedback while a save operation is in progress.
- **Save successful:** Display a concise confirmation.
- **Empty project list:** Explain that projects will appear when project data is available.

PAGE 5 — VISUAL DESIGN AND ACCESSIBILITY DECISIONS

1. Color System

The visual direction is premium, restrained and professional.

Token	Hex value	Intended use
Midnight Navy	#101B33	Main backgrounds and navigation
Deep Navy	#182844	Cards and secondary surfaces
White	#FFFFFF	Primary text on dark backgrounds
Soft White	#F5F7FB	Light surfaces and page backgrounds
Gold	#D4A853	Selected accents and highlights
Slate	#64748B	Secondary text on light surfaces
Success Green	#15803D	Successful actions and saved state
Error Red	#B91C1C	Validation and error messages

Color combinations must be checked for sufficient contrast before implementation. Gold is an accent rather than the default color for small text on white backgrounds. Status indicators also use text or icons so meaning is not communicated by color alone.

2. Typography

The application uses platform system fonts for performance, native rendering and offline availability.

Text role	Suggested size	Weight
Main display heading	36–44	Bold
Screen heading	30	Bold
Section heading	24	Semibold
Subheading	20	Semibold
Body text	16	Regular
Secondary text	14	Regular
Labels and captions	12–14	Medium or semibold

A consistent typography scale supports visual hierarchy. Long descriptions use comfortable line spacing, and the interface avoids excessive uppercase text or decorative font styles.

3. Spacing and Layout

The interface follows a spacing scale based on multiples of four density-independent units:

- 4–8 units for compact relationships.
- 12–16 units for related controls and card content.
- 20–24 units for screen padding and major sections.
- 32 units or more between distinct content groups when space permits.

Flexbox is used for responsive alignment. Cards can grow vertically when text scales. Controls should not overlap or become clipped on narrow Android screens.

4. Accessibility

- Text uses React Native's normal font-scaling behavior.
- Interactive controls have descriptive accessibility labels and appropriate roles.
- Buttons provide clear visible states and sufficiently large touch targets.
- Important information is not conveyed through color alone.
- Form errors are placed beside the relevant fields.
- Images have meaningful accessibility descriptions when they convey information.
- Keyboard focus and text entry remain usable during profile editing.
- Layouts are tested with larger Android system text settings.

5. Offline-First Design

The core portfolio content, editable profile and locally stored profile image should remain accessible in Airplane Mode. Optional repository and demonstration links are clearly labelled as online-only.

Local persistence uses AsyncStorage for low-risk profile information. Image references must point to files that remain available after an app restart.

6. Design-to-Implementation Verification

Before submission, the final screens should be compared with these wireframes. The Home identity sentence, verification code, project details, validation messages, media controls and restored profile state must be checked in the installed Android application.

Conclusion: CMPortfolio is designed to communicate personal identity and practical software-development ability through a consistent, accessible and offline-ready mobile experience. The final implementation and screenshots must reflect the actual features completed and tested.

Document status: Initial UI/UX design specification. Update the document with genuine final screenshots and any justified design changes after implementation.